

PYTHON MINI PROJECT REPORT

Project Title	Milling Operation MRR & Power Requirement Estimator	
Course Name	Semester-3: Python Programming	
Class Name	Mechanical-3B	
Branch Name	Mechanical Engineering / Automation & Robotics	
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Submitted to	Department of Mechanical Engineering, LJ Polytechnic, Lok Jagruti Kendra University, Ahmedabad, Gujarat, India.	

Certificate

This certificate is proudly presented to the following students for successfully completing the Python Mini-Project entitled “Milling Operation MRR & Power Requirement Estimator” as part of their Diploma in Mechanical Engineering curriculum. This project work has been carried out under my guidance and supervision and it is up to my satisfaction.

Name of Students	Enrolment Number	Class
ARMAAN SAIYED		Mechanical-3B
REHAN KHAN		Mechanical-3B
SAJJAD HUSSAIN		Mechanical-3B
OWAIS KHAN		Mechanical-3B

Date:	Name & Signature of Guide
Place: Ahmedabad	Mr. MohammedAzim Shaikh

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The Milling Operation MRR & Power Requirement Estimator is a Python-based engineering application developed to estimate important machining parameters for slab and face milling operations. The program accepts cutter diameter, number of teeth, feed per tooth, cutting speed, depth of cut, and width of cut as inputs. It calculates table feed rate, material removal rate (MRR), and estimated spindle motor power required.

1. Introduction

Briefly describe the purpose of your project. Mention the problem you are trying to solve and how it connects to mechanical engineering.

The objective of this project is to develop a simple and user-friendly Python tool for milling-process calculations. The estimator helps the user calculate table feed rate, material removal rate, and approximate spindle motor power requirement from standard milling input parameters.

2. Objective

Clearly define the objective of the project. State the problem or task that you are solving using Python.

The project uses Python to collect milling parameters from the user and apply standard machining relationships. The program calculates spindle speed from cutting speed and cutter diameter, calculates table feed from feed per tooth, number of teeth, and spindle speed, and then estimates MRR using table feed, depth of cut, and width of cut. The required spindle power is estimated from MRR and the material-specific machining power requirement. The interface can be implemented using Python libraries such as Tkinter for the graphical user interface and standard Python mathematical operations for the calculations.

3. Methodology

Discuss the methodology you followed to complete the project. Mention the algorithms, methods, or tools, libraries used in Python to achieve your objective.

This project applies directly to machining and manufacturing engineering. Milling operations remove material from a workpiece using a rotating multi-point cutting tool. Estimating feed rate, MRR, and spindle power helps in selecting suitable machining conditions and checking whether the available machine tool has adequate power for the operation. The estimator can be used for both slab and face milling calculations.

4. Mechanical Engineering Application

- Explain how this project applies to a mechanical engineering concept.
- Example: "The project uses Python to simulate beam deflection and calculate stress and strain in mechanical systems."

The Python program takes cutter diameter, number of teeth, feed per tooth, cutting speed, depth of cut, and width of cut as inputs. It then performs the following calculations:

1. Spindle speed, $N = (1000 \times V_c) / (\pi \times D)$
2. Table feed, $V_f = f_z \times z \times N$
3. Material removal rate, $MRR = V_f \times a_p \times a_e$
4. Estimated spindle power is calculated from the MRR and the selected specific cutting-power value.

The program displays the calculated values in a clear results section.

5. Python Code

Present the Python code used in the project.

(Ensure proper formatting, indentation, and comments for clarity.)

You can use the following tool to generate Code Snippets <https://carbon.now.sh/>

```
# Sample code block
def beam_solver(load, length):
    # Code to calculate bending moment
    return moment
```

Include screenshots of the Milling Operation MRR & Power Requirement Estimator running in Python, showing the input fields, calculated table feed rate, MRR, estimated spindle motor power, and any available visual representation of the milling operation.

6. Screenshots

Include screenshots of your Python program running, outputs, and any visual representations (graphs, charts, etc.).

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The estimator provides the calculated spindle speed, table feed rate, material removal rate, and estimated spindle motor power for the entered milling conditions. The results can be used to understand how cutter diameter, number of teeth, feed per tooth, cutting speed, depth of cut, and width of cut affect machining productivity and machine power requirements.

Higher feed, depth, or width of cut generally increases the material removal rate and therefore the estimated power requirement.

7. Results and Discussions

Explain the outcomes of your project. Discuss any graphs or outputs generated by the code. Provide interpretation in the context of mechanical engineering.

The Milling Operation MRR & Power Requirement Estimator demonstrates how Python can be applied to a practical mechanical-engineering machining problem. The project provides a structured method for calculating table feed rate, MRR, and estimated spindle motor power from basic milling parameters. It can help students understand the relationship between cutting conditions and machining requirements.

8. Conclusion

Summarize the overall findings. Discuss whether the project objectives were achieved and the limitations (if any).

1. Standard machining and milling-operation formulae used in manufacturing engineering.
2. Python programming documentation and standard mathematical operations.
3. Mechanical engineering manufacturing-process reference material.

9. References

List the references (books, research papers, online resources) used to gather information for the project.

10. Course Certificates
